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LawNet-Viz

A Web-based System to Visually Explore Networks of Law Article References

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What is LawNet-Viz?

- Web-based tool for the modeling, analysis and visualization of law reference networks
- Support to legal research tasks for referred and semantically related articles
- Interactive, fluid and effective user experience

Who is LawNet-Viz for?

Legal Experts

Identify relevant statutes in a cost- and time-effective manner.

Benefit in drafting new codes or abrogating existing ones, by inferring semantic relations between legal documents.

Laymen/Citizens

Legal aid to those who are not familiar with the legal domain.

Serve their search and consultation needs.

What is behind LawNet-Viz?

Network Science

Macro-, meso- and microscopic level analysis

Node centrality

Ranking measures

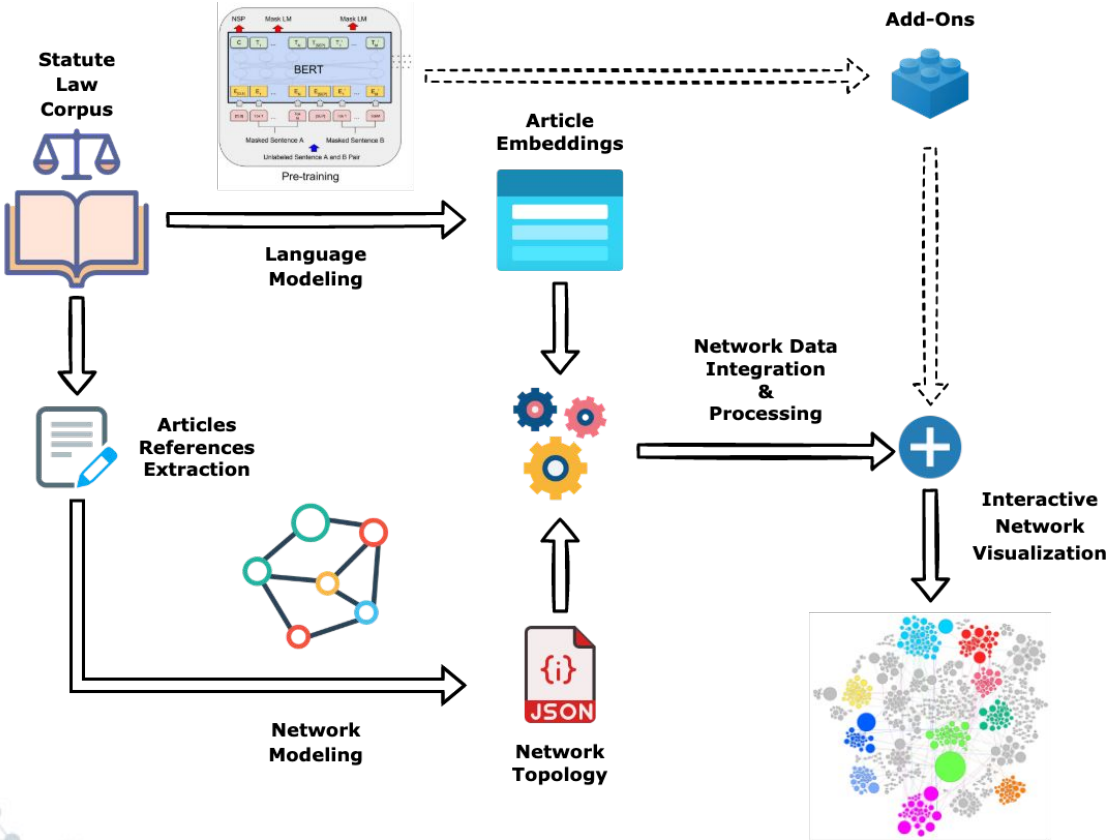
NLP

Sparse vectorial representations

Context-free embeddings

Deep contextualized embeddings

Workflow Overview



Node and Edge Features

Network Topology

Overall number of articles and references in the network, depending on the filtering threshold

Article Metadata

Article ID, Article title, Article text, Book ID, Cluster ID, Layout coordinates, color, size

Article References

Overall, incoming and outgoing references involving a target article

Centrality Scores

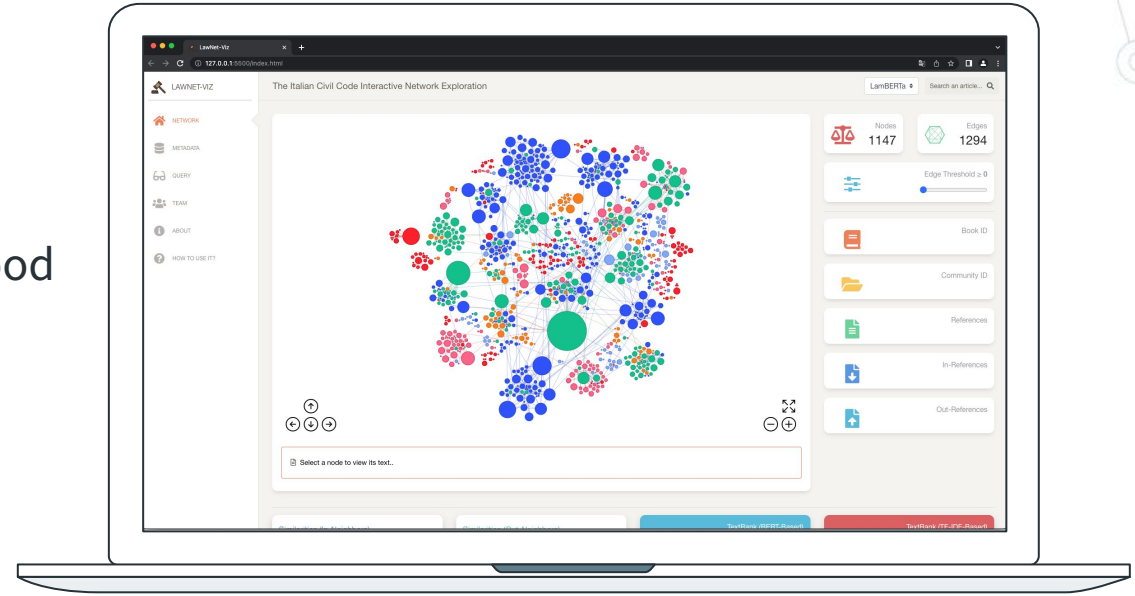
Betweenness, closeness, PageRank, Eigenvector centrality, and two variants of TextRank

Similarity Weights

Cosine similarity values of the vectorial representations (e.g., TF-IDF, BERT, etc.) of the articles

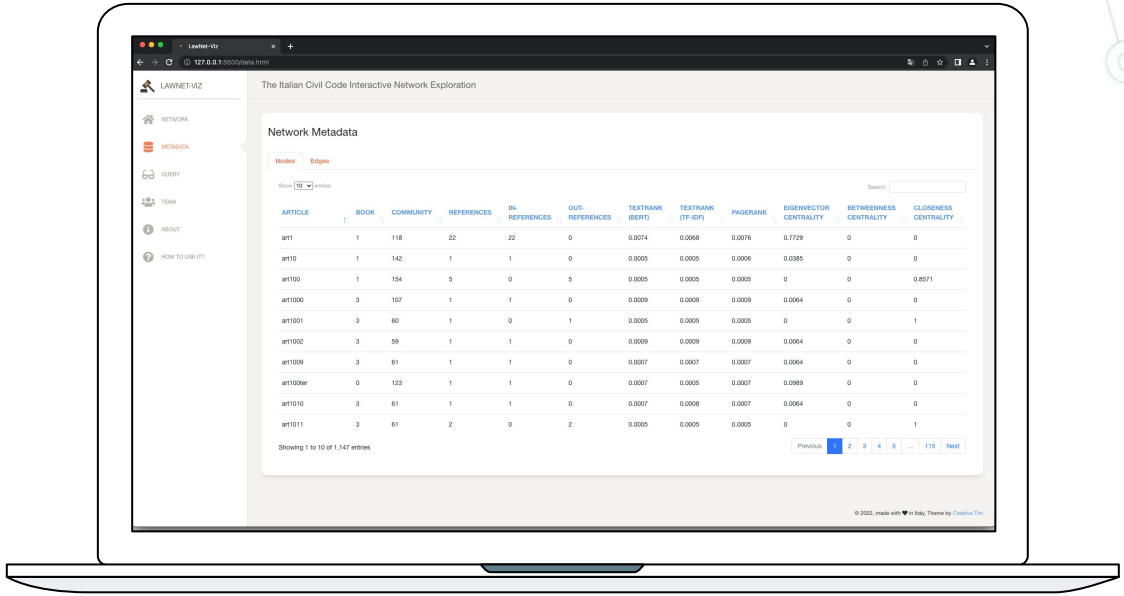
Interaction Modes - Network

- User-friendly exploration and manipulation of the article references network
- Articles' neighborhood inspection
- Topological and content statistics



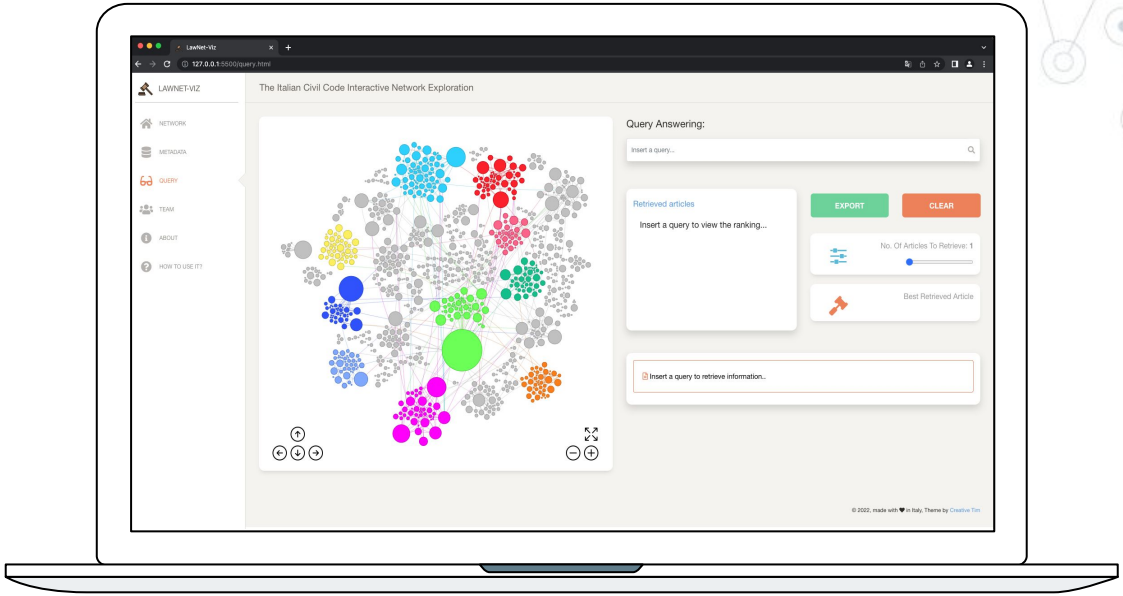
Interaction Modes - Metadata

- Interactive tabular exploration of the metadata associated with articles and references between articles
- Data filtering and sorting based on user needs



Interaction Modes - Query (*add-on*)

- Retrieval of articles relevant to a given input natural language query
- Selection of text representation model
  - sparse vectorial model
  - topic model
  - context-aware language model
- Reference-network positioning and content inspection of the results



LawNet-Viz in action: the Italian Civil Code (ICC)

Divided into 6 books, each on particular civil law theme:

- Book-1, on *Persons and the Family* (articles 1-455)
- Book-2, on *Successions* (articles 456-809)
- Book-3, on *Property* (articles 810-1172)
- Book-4, on *Obligations* (articles 1173-2059)
- Book-5, on *Labor* (articles 2060-2642)
- Book-6, on the *Protection of Rights* (articles 2643-2969)

	Book-1	Book-2	Book-3	Book-4	Book-5	Book-6
# articles	395	345	364	891	713	331
# articles w/ references	123	71	45	77	258	88
# article-references	243	132	70	118	551	180
avg. # article-references per-article	0.615	0.383	0.192	0.132	0.773	0.544
avg. # article-references per-article w/ references	1.976	1.859	1.556	1.532	2.136	2.045

(La Cava, Simeri, Tagarelli, 2021).
The Italian Civil Code Network Analysis.
Relations in the Legal Domain @ICAIL 2021 – The 18th International Conference on Artificial Intelligence and Law

Extending the capabilities of LawNet-Viz: interoperability with the LamBERTa framework

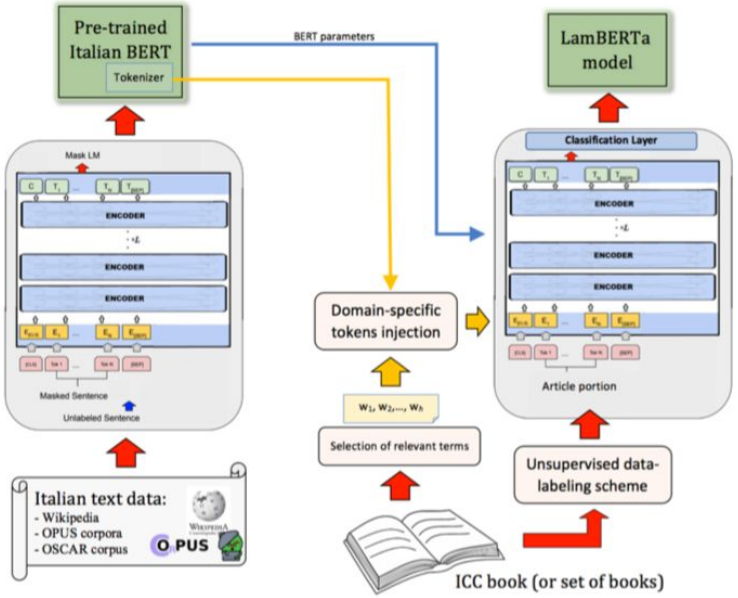
BERT-based framework for law article retrieval as a prediction problem

- Hundreds/thousands of classes
- Few-shot learning

Designed to fine-tune an Italian pre-trained BERT on a sequence classification task, given in input

- The whole ICC – *global* model
- A portion (i.e., book) of ICC - *local* model

Configurable w.r.t. unsupervised training-instance labeling scheme



(Tagarelli, Simeri, 2021). **Unsupervised law article mining based on deep pre-trained language representation models with application to the Italian civil code**. *Artificial Intelligence and Law,* pp 1-57, DOI: 10.1007/s10506-021-09301-8

Path to impact

- Enhance access to justice to legal professionals, government and administrative agencies
  - Reduce the workload
  - Improve productivity
  - Ensure efficiency and effectiveness
- Shrink the gap-to-justice involving citizens and laymen
  - Act as touch-point with a potentially intricate domain
  - Simplify access to regulatory systems

LawNet-Viz is currently planned for product development, with initial focus on the landscape of Italian and EU legal sources.

Thanks!

Any questions?

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